

NELSON KANG

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Education

University of Waterloo

Expected Apr 2030

Bachelor of Computing and Financial Management (CFM)

Activities: Competitive Intramural Volleyball Captain, Training for Ironman 70.3 Muskoka 2027

Awards: 1st of 18, Market Beat Portfolio Competition (2025)

Technical Skills

Languages: Python, C#, C, C++, SQL, TypeScript, JavaScript, HTML/CSS

Data & Infrastructure: AWS (S3, Athena, EventBridge, CodePipeline), Apache Iceberg, PyIceberg, DuckDB, Kafka, PostgreSQL, SQL Server, Redis, Docker, Terraform

Frameworks & Tools: FastAPI, Next.js, Celery, Modal, Supabase, Pandas, NumPy, Power BI, Postman, Git, Linux

ML / CV: Pytorch, YOLOv8, Roboflow, OpenCV, FFmpeg

Experience

AI Engineer (Part-time)

Sep 2026 – Present

Skrypt Health

Remote

- Building **lab-result triage** on a live EHR, routing abnormal results into a staff review queue. Designed an **autonomous coding harness**, and fixed a **production outage** that had hidden every clinic support request since launch.

Backend & Data Engineering Co-op

Jun 2026 – Aug 2026

Aquanow

Toronto, ON

- Migrated the company-wide month-end client reconciliation from a legacy JavaScript job to a containerized, **Kafka-triggered Lakeflow** worker (internal job framework) over **Apache Iceberg**, validating **3,800+** account identities against legacy output; root-caused **two long-standing production discrepancies**.
- Designed and built an **AI-agent reconciliation workflow** that classifies breaks by failure mode and auto-investigates unexplained residuals, producing review-ready reports for the finance team.
- Extracted a shared **reconciliation library**, reused it on a second reporting domain alongside a second co-op engineer, and authored the org-level **RFC and ADR** proposing it as a company-wide pattern.
- Diagnosed and fixed **5 critical production bugs** in a live regulatory reporting pipeline (**Python, AWS S3/Athena, PyIceberg**) submitting daily trading activity to **Turkey's securities depository (MKK)**, including a full-day snapshot-boundary offset and an in-asset fee misattribution; restored reconciliation from **0 to all 107 currencies**.

Software Engineer Intern

Jun 2024 – Aug 2024

Ericsson Canada Inc.

Ottawa, ON

- Built **4 C#/SQL Server ETL pipelines** from Jira into reporting tables with **temporal** change history, and audited every team pipeline into the reference that became the team's standard **debugging resource**.

Projects

ClipFarm — Automated Volleyball Highlight Generator

Mar 2026 – Present

Founded and lead a team of 4 · Python, FastAPI, Celery, Redis, Modal, TypeScript, YOLOv8-pose

- Turns a full match video into a filtered feed of **highlight clips**: a custom-trained ball detector bounds rallies by trajectory, cheer audio and rally-shape scoring keep the ones worth watching, and **YOLOv8-pose** labels each by action.
- Full stack: **FastAPI** endpoints with **Supabase JWKS** auth and **Row-Level Security**, **Celery + Redis** jobs, **Modal** GPU inference, and a **Next.js** clip grid whose correction UI feeds future training.

Dogwatch — Arctic Search Fleet Coordinator

Sep 2026

2nd Place, Dominion Dynamics WHITEOUT Track @ Hack the North 2026 (\$1,000) · Python, MAVLink, ArduPilot

- Four machines (quadcopter, fixed-wing, two towers) search a **25 km Arctic strait** for an unlit vessel and hold a track once found; a water-only **probability field** counts every empty camera sweep as evidence, so the fleet reasons instead of sweeping.
- Designed the **autonomous build harness** our two-person team ran on: planner, implementer, reviewer and fixer agents in isolated worktrees off a ranked backlog; **56 merged PRs** and **739 tests** in **29 hours**.

HUME — Belief-Revision Engine for Biology Research Findings

Jul 2026

1st Place, CORTEX BioSciences Track @ Hack the Six (\$1,500) · Python, no external libraries

- Reads a stream of biology findings and tracks which claims still hold, sizing each update in **log-odds** and scoping refutations to the condition that breaks them. **No LLM**, and an independent **90-attack red team** produced **zero** text-driven mutations.